Prime-field index calculus: the distance to rho in the lab's own observable, the equation budget every new lead must meet, and two cheap tests

2026-11-10 · `research/prime_ic_leads_20261110/` · crypto `main` f14b66786b · desk study, derivations and a development check; no S, no cost ratio, no leaderboard row; every measurement here is `[pcc:unverified]`

1. Answer

On the generic track parity with rho is impossible and the gap widens like $r^{1/6}$; its value is measured constant gains. On the algebraic track the only exponent lane on record, the PKM tower family, is converging to closure: the last fall degree d_F grows with the tower height at one degree per four levels (C-step holds at $t = 11, 12$; $t = 13$ reads ≥ 6) and with the arity at half a degree per unknown bit (stage C). Section 3 turns that growth into the distance to parity in the lab's own observable; section 4 names the mechanism as an *equation budget*: one cross-summand equation per relation over F_p, every further one costing about $\log_2 p$ unknown bits, against one per bit for Weil descent over F_{2^n}. Every idea examined today fails that budget (section 5); the one family that passes it for free, the $[k]$-multiple consequences of the relation, is implemented as a presentation control with pre-stated predictions (section 6). Nothing here reaches parity and nothing on record can; the note states exactly what a new lead would have to supply.

2. Where the prime-field program stands (measured, by path)

item	value	source
Generic track, Table B, `icv1-fp10`…`fp24`	IC / rho-negation 5.47× to 13.5×	`docs/ic/LEADERBOARD.md` Table B
20-bit factor base $N = 32 \to 45$	gap 9.47× ⟶ 7.75× [6.92, 8.70]; second-host replay 1,008/1,008	`research/prime_base_oracle_tuning_20261029/`
Algebraic-track prime baseline	empty (R1 Option C)	`research/prime_ic_program_20261029/DECISIONS.md`
`pkm-semaev` smoke, Curve25519 miniature $k = 20$, $t = 2$	S = 1027 / 49,513 / 485,893 at w = 0 / 0.1 / 1 vs 9.05 / 9.46 / 13.16 generic	`problems/curve25519-family-algebraic-ic-tower.md`
PKM Kummer $m = 2$, P_1, d_F by t	3, 4, 4, 4, 4, 5, 5, 5, 5, 6 ($t \leq 10$); 6 ($t = 11, 12$); ≥ 6 ($t = 13, 14$)	`research/pkm_falldegree_unblock_20261107/RESULTS.md`, `RESULTS_T12_BRACKET.md`
Stage C, fixed t, growing m	$t = 2$: 4, 5, 6, 6, ≥ 7; $t = 3$: 4, 6, 7, ≥ 8, ≥ 8 at $m = 2..6$; "grows in m" on all six ladders	`RESULTS_STAGE_C.md`
Exponent study, symmetrised subgroup base	$m = 2$: $s = 2.77$ [2.31, 3.23] at $k = 20$; $m = 3, 4$ undetermined (censored)	`research/curve25519_exponent_study_20261108/`
Decisive cells	$t = 14$ cap 7 queued; $m = 4$, $N = 20$ closure not launched (112 GiB, 49 h, Spot)	`docs/session-logs/2026-11-10.md`

3. What parity requires (derived)

With base size B, arity m, per-target cost $T = B^s$ and a sparse solve $\approx mB^2$, the total is about $r^{2/(m+1-s)}$, below $r^{1/2}$ iff $s < m - 3$ (program note §2.4; audit `semaev_alternatives_audit_20261104`). A dense closure or Macaulay elimination reaching degree $d = \alpha N$ in $N = m \log_2 B$ one-bit unknowns meets $M(N, d) = \sum_{k \leq d} \binom{N}{k} \approx 2^{NH(\alpha)}$ columns, so $s \geq mH(\alpha)\omega'$ and the lane can reach parity only if $H(\alpha) < (1 - 3/m)/\omega'$:

m	$\alpha^*, \omega' = 1$	$\alpha^*, \omega' = 2$	$\alpha^*, \omega' = 3$
4	0.042	0.017	0.010
5	0.079	0.031	0.019
6	0.110	0.042	0.025
8	0.156	0.056	0.033
10	0.189	0.066	0.038

Measured slopes: $1/8 = 0.125$ per bit in t (C-step, $m = 2$) and 0.5 per bit in m (stage C, $t = 2$ and $t = 3$). Both exceed every threshold at $\omega' \geq 2$; the m-slope exceeds every threshold at $\omega' = 1$ too. Recorded `max_cols` = 16,572 at $t = 13$, cap 5, against $M(26, 5) = 83,682$: F4′s matrices are about 5× sparser than the dense count, a constant that does not move s. This is a lower

bound for dense degree-d elimination and a model for F4; it assumes the slope persists beyond $N \leq 28$, which the $m = 4$, $N = 20$ closure tests.

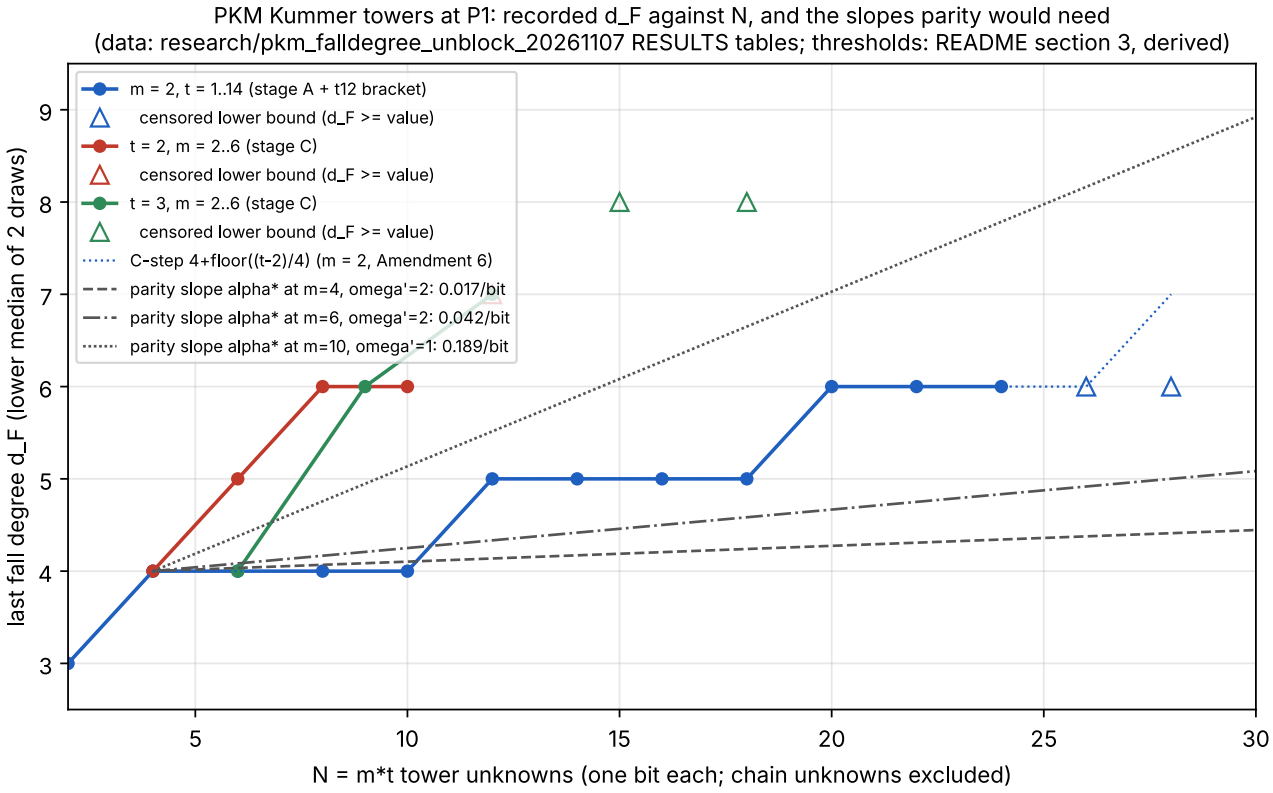


Figure 1: Recorded d_F ladders (lower medians of 2 draws; triangles are censored lower bounds) against N, with the C-step prediction and the α^* parity slopes of section 3. Data: `research/pkm_falldegree_unblock_20261107` RESULTS tables; thresholds derived here.

4. The cross-summand equation budget (derived)

Weil descent over F_{2^n}: one relation gives n equations of degree 2 in $mn' \approx n$ bit unknowns. A PKM tower over F_p: $t = \log_2 B$ one-bit unknowns per summand and **one** equation, with no Galois group to split it. A second equation from the same relation costs about $\log_2 p$ unknown bits in every route: a fixed rational image $\varphi(P)$ in its own tower ($|V| \cdot |V'| = Bp$), the extension F_{p^k} (Gaudry cost $p^{2-2/k}$), a p-adic digit (the torsion lift's second digit has degree about N^2), or a jet component. The only free family is $[k]R = \sum [k]P_i$, which enters the Kummer quotient at degree about $2\log_2 k + 2$ (6 for $k = 2$). Hence a tower system is a one-generator system in a 2^N-point quotient ring and its fall degree grows. The filter for any new lead: equations per unknown bit after the membership equations are removed; whether every extra equation brings a new F_p unknown or shrinks the base; the degree at which consequence equations enter.

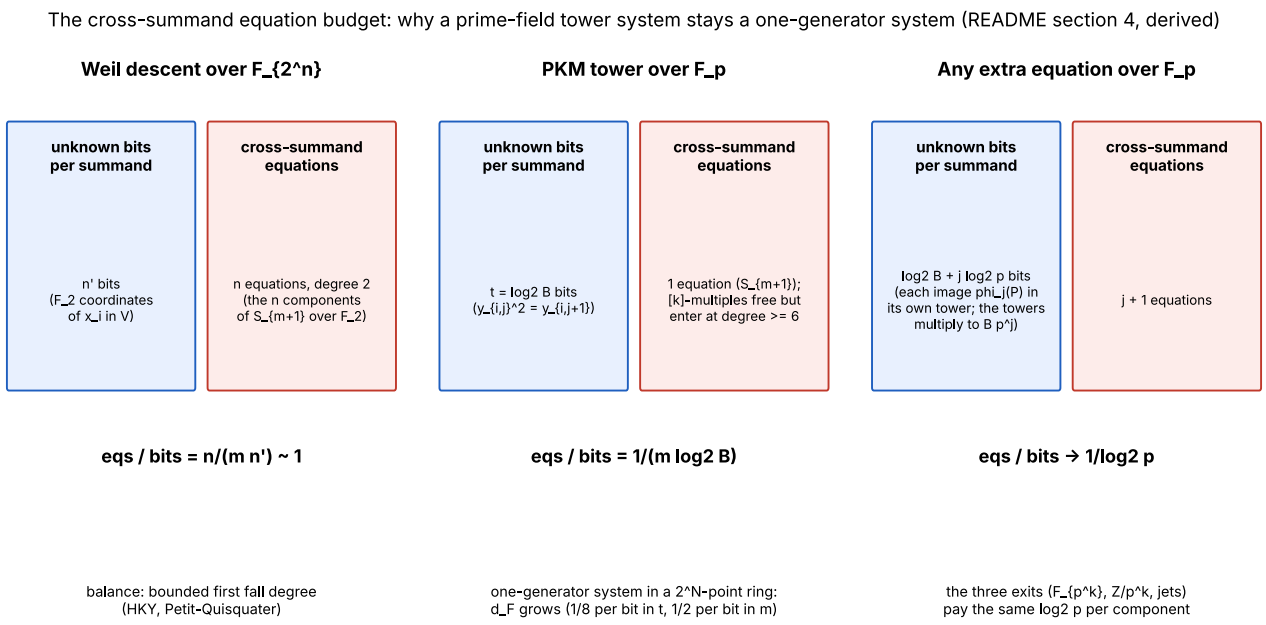


Figure 2: The equation budget: Weil descent (balance), the PKM tower (one generator), any extra equation over F_p (ratio $1/\log_2 p$).

5. Examined today and closed on paper

2-torsion (T_0) symmetrised tower on Montgomery-form curves: the 2-isogeny with kernel $(0,0)$ has x-map $x \mapsto x + A + B/x$, so a tower on the invariant is a plain tower on $E/\langle T_0 \rangle$, same one-generator system (constant column gain only; Kummer unavailable on Curve25519, $v_2(p-1) = 2$). Translate-pair and endomorphism-image bases: $|V| \cdot |V'| = pB$, the second equation costs $\log_2 p$ bits. Wagner k-tree filtering on tower sets: needs $|F| \geq \sqrt{2p}$. p-adic second digit of the torsion lift: degree N^2, or a free F_p unknown per summand. For $j = 0$, the ω-conjugate equation is the original up to relabelling ($S_3(\zeta x) = \zeta S_3(x)$). Atlas matches are listed in the README.

6. The multiples presentation control

`examples/pkm_tower_pilot.rs --controls multiples` appends to the tower system the cleared $S_{m+1}(x([2]P_1), \ldots, x([2]P_m), x([2]R)) = 0$, with $x([2]P) = N(x)/D(x)$, $N = x^4 - 2ax^2 - 8bx + a^2$, $D = 4(x^3 + ax + b)$ (for chains, doubled partial-sum unknowns linked by $D(u_k)u'_k - N(u_k) = 0$). Same cell seed, same tower, curve and targets as the tower control; every appended equation vanishes at planted solutions (asserted; 12/12 unit tests). It enters at multilinear degree 6, so at $m = 2$ the two presentations must agree wherever $d_F \leq 5$ (P1, a built-in control); at cells with $d_F(\text{tower}) \geq 7$ the prediction was $d_F(\text{multiples}) \in \{d_F - 1, d_F\}$ with the slopes unchanged (P2). Falsifier: a drop of at least 2 at a cell with $d_F(\text{tower}) \geq 8$, or an m-ladder that reads flat.

Development check (this host, closure engine, Kummer at P_1, seed `0x504B4D54`, two random targets per cell, both arms on the same targets; no isolation; every value `[pcc:unverified]`; rows and table in `devcheck/`). The tower arm reproduces stage A's medians at $t = 2..9$ and stage C's cells, the instrument check.

cell	d_F tower	d_F multiples	Δ	falls tower / multiples
$m = 2, t = 2..9$ (16 rows)	4, 4, 4, 4, 5, 5, 5, 5	identical on every row	0	identical
$m = 2, t = 10$	> 6 (budget-censored)	> 6 (same)	—	[4, 5] / [4, 5]
$m = 3, t = 2$ ($N = 6$)	5, 5	5, 5	0	[4, 5] / [4, 5]
$m = 3, t = 3$ ($N = 9$)	6, 6	5, 5	**−1**	[4, 5, 6] / [4, 5]
$m = 4, t = 2$ ($N = 8$)	6, 6	5, 5	**−1**	[4, 5, 6] / [4, 5]
$m = 4, t = 3$ ($N = 12$)	7, 7	6, 6	**−1**	[4, 5, 6, 7] / [4, 5, 6]
$m = 5, t = 2$ ($N = 10$)	6, 6	6, 6	0	[4, 5, 6] / [4, 5, 6]
$m = 5, t = 3$ ($N = 15$)	> 7 both draws ($c = 8$ budget-censored)	> 6 both draws ($c = 7$ censored by the entry cap at 399,850 columns)	inconclusive	[4, 5, 6, 7] / [4, 5, 6]

P1 holds (identical at $m = 2$, draw by draw). P2 holds and nothing reaches the falsifier: at $m \geq 3$ the drop is exactly one degree on six of ten completed cells and zero on four, and both slopes are unchanged ($m = 4$: 6 ⟶ 7 against 5 ⟶ 6 from $t = 2$ to 3; $t = 3$: 6 ⟶ 7 against 5 ⟶ 6 from $m = 3$ to 4). The one-degree offset at $m \geq 3$ arrives with the degree-4 doubling links and the extra unknowns u'_k, a presentation shift, not a change in growth. The $m = 5$, $t = 3$ cell is censored on both arms (tower > 7, multiples > 6) and is inconclusive on this host. Section 4 item 3 stays closed as far as this check can test it; not tested: [3]- and [4]-multiples together, larger N, an F4 run on the augmented systems.

7. Null-support slope (lead 3, development check, 2026-11-11)

Pre-registered before the rows existed. The `null` control replaces every summation equation by a random polynomial on a chosen support, with the same tower and targets as the `tower` arm: S_3's own support, or the full exponent box $0..W$ (`--null-width`). Every value `[pcc:unverified]`; rows in `devcheck/`.

support	$m = 2, t = 2..8$	$m = 3, t = 2, 3$	$m = 4, t = 2, 3$
tower (S_3, the curve)	4, 4, 4, 4, 5, 5, 5	5, 6	6, 7
null on S_3's support	4, 4, 4, 4, 5, 5, 5	5, 6	6, 7
box $W = 1$	3, 3, 4, 4, 4, 4, 5	4, 4	4, 5
box $W = 2$	4, 4, 4, 4, 5, 5, 5	5, 6	6, 7
box $W = 3$	5, 5, 5, 5, 5, 5, 5	8, 8	> 8
box $W = 4$	5, 5, 5, 5, 5, 5, 5	> 8	> 8

Shape-driven, as predicted: a random polynomial on S_3's support or on the $W = 2$ box has the tower's d_F on all 40 completed cells, draw by draw, with the same fall sequences. The degree is monotone in the support width (at $m \geq 3$ the $W \geq 3$ boxes also raise the degree in the chain's free unknowns, so those rows are reported, not read). The curve contributes nothing the support does not: the growth is a property of the carry-graded one-generator closure, the mechanism of section 4, and the remaining step on this lead is the derivation of the $d_F(W, N)$ law, a computation without any curve.

8. Leads, ranked

1. Finish the decisive PKM cells: $t = 14$ cap 7; the $m = 4$, $N = 20$ closure (Amendment 8 readings).
2. The multiples presentation probe: done (section 6): P1 and P2 hold, the falsifier is not reached, one cell inconclusive.
3. The null-support slope: measured (section above): curve-independent on all 40 cells; what remains is the derivation of $d_F(W, N)$.
4. The first counted algebraic-track row: specification written (`algebraic_row_spec/`: 3 Table B curves × 8 targets × 14 arms on the isogeny base, protocol draft validated, derived 46 vCPU-hour ceiling inside the approved budget); the owner decides the submission route; it needs an EKS identity and an image rebuild.
5. Generic-track constants: the prefetch Design A patch is obsolete (reconciled 2026-11-11: `layout=lines` is already on `main` and measured 12 % slower at fp28 in the v6 tournament, `invalid_measurement`); what stays open is the software-prefetch step on the landed layout and $N > 45$ at 20 bits: the only place a measured improvement over the current baseline is available.
6. Honest non-uniform accounting (ST^2 per row; preprocessing-rho reference).

What would change the list: an equation source that passes the filter of section 4. None is known. On 2026-11-11 the jet and Z/p^k exits were derived and measured against the filter (`jet_exit/`): dual numbers add only free F_p unknowns, and the torsion section's p-adic digits are generic on Kummer bases (popcount degree equal to random and to the full-rank degree, 3 to 6 for $t = 6$ to 13, on 42 cells over 17 curves, two primes and five integer lifts), so all three bigger-ring exits are closed at toy scale ($t \leq 13$, 14- and 16-bit primes; the filter's $\Theta(N)$ is a requirement, not a measured asymptotic). Canonical (CM) lifts of two class-number-one curves and the Dickson base read the generic degree as well (`jet_exit/` section 7). The atlas delta recording this is `jet_exit/ATLAS_DELTA.patch`; the atlas dispositions themselves (two `open_next_step`, one `closed_obstructed`) are the reviewer's to change.

9. Sources checked

Knowledge base unavailable (`research-kb` MCP absent; `scripts/research-kb status`: no executable); fallback: the lead atlas (`crypto-autoresearcher`, 176 leads, 30 in `ic-prime/`, `--match` run per idea) and the repository files listed in README §0. Problem record: `problems/prime-field-cross-summand-equation-budget.md`.